

High Score

Time limit: 1.00 s

Memory limit: 512 MB

You play a game consisting of n rooms and m tunnels. Your initial score is 0, and each tunnel increases your score by x where x may be both positive or negative. You may go through a tunnel several times.

Your task is to walk from room 1 to room n . What is the maximum score you can get?

Input

The first input line has two integers n and m : the number of rooms and tunnels. The rooms are numbered 1, 2, ..., n .

Then, there are m lines describing the tunnels. Each line has three integers a , b and x : the tunnel starts at room a , ends at room b , and it increases your score by x . All tunnels are one-way tunnels.

You can assume that it is possible to get from room 1 to room n .

Output

Print one integer: the maximum score you can get. However, if you can get an arbitrarily large score, print -1 .

Constraints

- $1 \leq n \leq 2500$
- $1 \leq m \leq 5000$
- $1 \leq a, b \leq n$
- $-10^9 \leq x \leq 10^9$

Example

Input	Output
4 5 1 2 3 2 4 -1 1 3 -2 3 4 7 1 4 4	5